Title of the Notes

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Just practice problems for Topology, Commutative Algebra, and perhaps Machine Learning and Advanced Algorithms.

1 Exam style stuff

I have to practice for MAT224 Commutative Algebra oral examiniation so these problems are more about intuition, allowing less rigor.

Problem 1.1 (Large algebra identification problem)**.** Let

$$R = k[x, y],\ I = (x^2, xy, y^2),\ A = R/I.$$

What can you say about A? In particular, is it Noetherian or Artinian? Can you give a k-basis or normal form for its elements?
Compute $\sqrt{I}$. Then describe $V(I) \subseteq k^2$. What information about I is visible from $V(I)$?
Is I prime? Is I primary? Give the quickest convincing argument you can.
Determine the maximal ideals of A. Is A local? If so, identify the maximal ideal $\mathfrak{m}$, and compute enough powers of $\mathfrak{m}$ to see what happens.
Compute $\mathrm{Ann}_A(\bar{x})$ and $\mathrm{Ann}_A(\bar{y})$. Are either of these prime ideals?
Consider mutliplication by $\bar{x}$,

$$\mu_{\bar{x}} : A \to A,\ a \mapsto \bar{x}a.$$

Without being given a sequence, identify $\ker \mu_{\bar{x}}$, $\mathrm{im}\, \mu_{\bar{x}}$, $\mathrm{coker}\, \mu_{\bar{x}}$.

Solution. We have $I = (x, y)^2$. We can then say that the ring is Noetherian by using Hilbert's Basis Theorem twice $k \to k[x] \to k[x][y] = k[x, y]$ with Noetherian preserved. Then $I \subseteq R$ is Noetherian so $R/I = A$ is Noetherian.

Artinian does not hold for R since $(x) \supsetneq (x^2) \supsetneq (x^4) \supsetneq \cdots$ never stabilizes. We have an exact sequence

$$0 \to (x, y)(x, y) \to R = k[x, y] \to A \to 0.$$

We have that R is Artinian if and only if $(x, y)^2$ and A are Artinian. We know that R is not Artinian, and $(x, y)^2$ is not Artinian either so we can't conclude anything about A using this.

What we can notice is that any polynomials with x degree ≥ 2 vanish, same for y, while terms in both x and y vanish also. Thus all polynomials have only constant terms, terms in x and terms in y. Thus A is an R-algebra, more specifically a k-vector space with basis $\{1, \bar{x}, \bar{y}\}$.

The root $\sqrt{I}$ is (x, y). We obviously have $(x, y) \subseteq \sqrt{I}$. For $ax + by \in (x, y)$ we have

$$(ax + by)^2 = a^2x^2 + (2ab)xy + b^2y^2$$

which is in I. Now, all generators of I are divisible by by either x or y so $I \subseteq (x, y)$. We have (x, y) prime (and even maximal) so $\sqrt{I} \subseteq (x, y)$ (since $f^n \in I \subseteq J$ implies $f \in J$ due to primality).

We do not have that I is prime, since x^2 is in I, but x is not. It is, however, maximal. We have already established that (x, y) is maximal and $I = (x, y)^2$ so

$$(x, y)^2 \subseteq I \subseteq (x, y).$$

Thus I is (x, y)-primal.

The maximal ideals of A map one to one with the maximal ideals of R containing I.

As mentioned we have $(x, y) \supseteq (x^2, xy, y^2)$ and (x, y) is maximal so

$$(\bar{x}, \bar{y}) = \frac{(x, y)}{(x^2, xy, y^2)}.$$

We can use the third isomorphism theorem to prove that $A/(\bar{x}, \bar{y}) \cong k$.

For uniqueness, suppose $f \in A$. Then $f = a + bx + cy$. The ideal $\mathfrak{m} = (\bar{x}, \bar{y})$ consists of all elements where the constant term is zero, $a = 0$, so an element f not in $(\bar{x}, \bar{y})$ is just $f = a \in k \setminus \{0\}$, which is invertible since k is a field.

As we glimpsed earlier, $\mathfrak{m}^2 = (\bar{x}, \bar{y})(\bar{x}, \bar{y}) = (\bar{x}^2, \bar{x}\bar{y}, \bar{y}^2) = 0$.

We have

$$\text{Ann}_A(\bar{x}) = \{a \in A \mid a\bar{x} = 0.\}$$

Equivalently, in $A = R/I$ we have that the annihilator of $\bar{x}$ are all $a = \alpha + \beta\bar{x} + \delta\bar{y}$ such that

$$a\bar{x} = \alpha\bar{x} + \underbrace{\beta\bar{x}^2}_{0} + \underbrace{\delta\bar{y}\bar{x}}_{0} = 0.$$

We have that $\bar{x}$ is a basiselement of the space, requiring $\alpha = 0$. β, δ can be any elements at all so

$$a = 0 + \beta\bar{x} + \delta\bar{y} \in (\bar{x}, \bar{y}) = \mathfrak{m}.$$

It is not terribly surprising that $\text{Ann}_A(\bar{x})$ is contained in $\mathfrak{m}$, since it is maximal, but actually $\text{Ann}_A(\bar{x}) = \mathfrak{m}$.

We can see this since, for $a\bar{x} + b\bar{y} \in (\bar{x}, \bar{y})$ we have

$$(a\bar{x} + b\bar{y})\bar{x} = a\bar{x}^2 + b\bar{y}^2 = 0.$$

Thus $\mathrm{Ann}_A(\bar{x}) = \mathfrak{m}$, the maximal ideal.

Now, the exact same holds for $\bar{y}$, for the same reasons, and it is not hard to see.

Now for multiplication at $\bar{x}$ we get a natural short exact sequence

$$0 \to A \xrightarrow{\mu_{\bar{x}}} A \xrightarrow{\pi} \mathrm{Ann}_A(\bar{x}) \to 0$$

■

Problem (Tensor products and quotient modules)**.** Let $R = k[x, y]$, $I = (x^2, xy)$, and $J = (y)$. Put $M = R/I$ and $N = R/J$.
Consider the R-linear map

$$\begin{aligned} \varphi : R &\longrightarrow M \otimes_R N, \\ r &\longmapsto (r + I) \otimes (1 + J). \end{aligned}$$

a) Prove that φ is surjective.
b) Prove that $I + J \subseteq \ker \varphi$.
c) Determine whether the induced map

$$\bar{\varphi} : R/(I + J) \longrightarrow M \otimes_R N$$

is an isomorphism.
d) Determine $M \otimes_R N$ explicitly and exhibit a k-vector-space basis.

Proof of a). Take $\bar{m} \otimes \bar{n} \in M \otimes_R N$. Then there is at least one each of m_0, n_0 such that $\bar{m} = m_0 + I$ and $\bar{n} = n_0 + J$.

Take $m_0 n_0 \in R$. Then

$$m_0 n_0 \mapsto (m_0 n_0) + I \otimes 1 + J = m_0 + I \otimes n_0 + J = \bar{m} \otimes \bar{n}.$$

■

Proof of b). Let $f \in I + J$. Then

$$f = ax^2 + bxy + cy, \ a, b, c \in k[x, y].$$

Then

$$f \mapsto (ax^2 + bxy + cy) + I \otimes 1 + J.$$

Then

$$(\underbrace{ax^2 + bxy}_{\bar{0}} + cy) + I \otimes 1 + J = cy + I \otimes 1 + J = I \otimes J = 0.$$

■

Proof of c). Intuitively, φ is surjective with $\ker\varphi = I + J$ (I think) so we know, by the first isomorphism theorem, that

$$R/(I+J) \cong M \otimes_R N.$$

The induced map $\bar{\varphi}$ sends $\bar{r} = r + (I+J)$ to

$$\bar{r} \mapsto \bar{r} + I \otimes 1 + J.$$

As mentioned, $\ker\varphi = I + J$ which corresponds to 0 in $R/(I+J)$. The induced map retains surjectivity as we simply take

$$\bar{m}\bar{n} \mapsto \bar{m} \otimes \bar{n}.$$

■

Solution to d). We have

$$M \otimes_R N \cong R/(I+J) = R/((x^2, xy) + (y)).$$

We have

$$(x^2, xy) + (y) = (x^2, xy, y)$$

so

$$R/(I+J) \cong k[x]/(x^2).$$

We then have a basis $\{1, x\}$ for $k[x]/(x^2)$ which corresponds to the basis

$$\{\bar{1}, \bar{x}\}.$$

■

Extra.

1.

Let

$$\begin{aligned} \beta : (R/I) \times (R/J) &\longrightarrow R/(I+J), \\ (r+I, s+J) &\longmapsto rs + (I+J). \end{aligned}$$

We show that β is well-defined.

Suppose $\bar{r} = r_0 + I = r_1 + I$ with $r_0 \neq r_1$.

Then

$$\begin{aligned}\beta(r_0 + I, s + J) &= (r_0 s) + (I + J) \\ &= \overline{r_0 s} \\ &= \bar{r_0}\bar{s} \\ &= \bar{r_1}\bar{s} \\ &= (r_1 s) + (I + J) \\ &= \beta(r_1 + I, s + J).\end{aligned}$$

The point is that

$$\bar{x}\bar{y} = \overline{xy}.$$

The argument is analogous for $\bar{s}$.

2.

For bilinearity, fix first s, supressing the equivalence class notation for now.

Then

$$\begin{aligned}\beta(r_0(r_1 + r_2), s) &= ((r_0(r_1 + r_2))s) \\ &= r_0(r_0 s + r_1 s) \\ &= r_0(\beta(r_0, s) + \beta(r_1, s)).\end{aligned}$$

We could also be more rigorous than this. The argument for fixed r is the same.

Since β is an R-bilinear function we invoke the tensor products unviersal property, giving a map $\hat{\beta} : R/I \otimes_R R/J \to R/(I + J)$ such that $\beta \circ ((r, s) \mapsto r \otimes s) = \hat{\beta}$. ■

Problem (Compactness and separation axioms)**.** Let X be a compact Hausdorff space.

a) Let $x \in X$, and let $F \subseteq X$ be closed with $x \notin F$. Prove that there exist disjoint open sets $U, V \subseteq X$ such that

$$x \in U, \qquad F \subseteq V.$$

b) Prove that every compact Hausdorff space is normal. That is, if $A, B \subseteq X$ are disjoint closed sets, there exist disjoint open sets $U, V \subseteq X$ with

$$A \subseteq U, \qquad B \subseteq V.$$

c) Let X be an infinite set equipped with the cofinite topology

$$\mathcal{T} = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}.$$

Determine, with proofs, whether X is compact, T_1, Hausdorff, and normal.

Proof of a). This follows from compact Hausdorff $\Rightarrow$ normal. $\{x\}$ is closed, since X is Hausdorff, so since X is normal, we can separate

$$\{x\},\ F$$

into disjoint open sets containing them.

Below will be a proof from first principles.

For $y \in F$ we can, by Hausdorffness find

$$U_y \ni x$$

and

$$V_y \ni y$$

disjoint and open.

We have

$$F \subseteq \bigcup_{y \in F} V_y.$$

Since X is compact we have

$$F \subseteq \bigcup_{i=1}^{n} V_{y_i}.$$

Set

$$V = \bigcup_{i=1}^{n} V_{y_i}$$

is open. For x, let U be

$$U = \bigcap_{i=1}^{n} U_{y_i}.$$

This is a finite intersection of open sets so open.

Now

$$U \cap V = \bigcup_{j=1}^{n} \left(\left(\bigcap_{i=1}^{n} U_{y_i} \right) \cap V_{y_j} \right).$$

We have

$$\left(\bigcap_{i=1}^{n} U_{y_i} \right) \cap V_{y_j} \subseteq U_{y_j} \cap V_{y_j}$$

so each term is empty.

Thus $U \cap V = \emptyset$.

■

Proof of (b). We proceed as above. If X is Hausdorff it is T_1.

For fixed $x \in A$ construct

$$V_x = \bigcup_{j=1}^{n} V_{y_i}$$

and

$$U_x = \bigcap_{i=1}^{n} U_{y_i}$$

as in (a) with the property that $U_x \cap V_x$

We do this for all $x \in A$. Then

$$\bigcup_{x \in A} U_x$$

covers A and can be reduced to

$$\bigcup_{i=1}^{n} U_{x_i}.$$

Corresponding V_{x_i}'s independently cover B, so their intersection also does, and it is disjoint to each U_{x_i} and thus disjoint to their union.

■

Proof of (c). It is indeed T_1 since

$$\{x\} = X \setminus (X \setminus \{x\})$$

meaning

$$X \setminus \{x\}$$

is open.

Take two distinct $x, y \in X$. Suppose $x \in U,\ y \in V$ are disjoint open sets.

Then

$$|X \setminus U| = n,\ |X \setminus V| = m$$

for nonzero integers m, n.

Since $U \cap V = \emptyset$ we have

$$U^c \cup V^c = X.$$

Then

$$X \cap (U^c \cup V^c) = X,$$

but then

$$|X| = |X \cap (U^c \cup V^c)| = |X \cap (U^c) \cup X \cap (V^c)| \leq n + m,$$

giving

$$n + m \geq |X| \geq \aleph_0,$$

a clear contradiction.

Thus X is not Hausdorff.

Let $\mathcal{U} = \{U_\alpha\}$ be a cover of some U. Pick some $U_1 \in \mathcal{U}$. Then $X \setminus U_1 = \{x_1, \dots, x_n\}$. For each x_i, if $x_i \in U$, take one of the U_α which contains it. Repeat until all x_i are accounted for. Then we have a cover with at most $n+1$ elements.

The space is not Hausdorff and thus not normal. Suppse it was normal. We know the space is T_1 so $\{x\}, \{y\}$ are closed. Then we can partition them into open disjoint sets, implying Hausdorffness. No good. ■

Problem (Integral extensions)**.** Let k be a field, and consider

$$A = k[t^2, t^3] \subseteq B = k[t].$$

a) State the definition of integrality and determine whether t is integral over A.
b) Show that every element of B has the form $a + bt$, where $a, b \in A$. Deduce that B is a finite A-module.
c) Prove that
$$\text{Frac}(A) = \text{Frac}(B) = k(t).$$
d) Determine whether A is integrally closed in its fraction field.
e) Determine whether every element of B is integral over A, and justify your conclusion.

Proof of (a). Suppose $R \subseteq R'$. $a \in R'$ is integral over R if it is the zero of a monic polynomial

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 x^0, \ a_i \in R.$$

If all elements in R' are monic over R, then R' is integral over R. More precisely, if $R' = R[c_1, c_2, \dots, c_n]$ then R' is an integral extension of R if all c_i are integral over R. The previous is also true since all elements of R are integral over R, take

$$x - a$$

for $a \in R$.

Take

$$f(T) = T^3 - t^3.$$

Then $f(t) = 0$ and

$$f(T) \in k[t^2, t^3][T]$$

so B is integral over A with generating set $\{1, t\}$. ■

Proof of (b). We want to show

$$B = A + At.$$

We have $t^2, t^3 \in A$ so we have

$$t^n \in A$$

for every $n \geq 2$. We could prove this inductively. The base case is trivial. Thereon out,

even powers are easy to see. For $n = 2k$ we have

$$(t^2)^k = t^{2k} = t^n \in A$$

For $n = 2k+1$ notice that $2k+1-3 = 2(k-1)$. If $k \geq 4$, then t^{2k-2} exists and thus

$$t^{2k-2}t^3 = t^{2k+1} = t^n \in A$$

exists. So after observing that $t^4 = (t^2)^2 \in A$ this follows.

■

Proof of (c). The forward inclusion $k[t^2, t^3] \subseteq k[t] \subseteq k(t)$ is trivial.

We have that

$$\frac{t^3}{t^2}\frac{t^2}{t^3} = 1,$$

but

$$\frac{t^2}{t^3} = \frac{1}{t}$$

so $(t^3/t^2)^{-1} = t^{-1}$, hence $t^3/t^2 = t$, meaning $k(t) \subseteq \mathrm{Frac}(k[t^2, t^3])$. ■

Problem (Ring, ideal, and module operations)**.** Let $A = k[t^2, t^3] \subseteq B = k[t]$.

a) Determine whether t^5, t^7, t belong to A.
b) Describe explicitly the ideals

$$I = (t^2, t^3)_A, \qquad J = (t^2, t^3)_B.$$

Explain why they are different.
c) Determine A/I and B/J, up to ring isomorphism.
d) Prove that B is finitely generated as an A-module, but that $\{1, t\}$ is not an A-module basis.

Proof of (a). We have that $t^n \in A$ for all $n \geq 2$, so $t^5, t^7 \in A$.

We can prove this inductively by observing that $t^2, t^3, t^4 = (t^2)^2 \in A$.

For even $n = 2k$ it is trivial as

$$(t^2)^k = t^n \in A.$$

By assumption, supposing $n = 2k+1-3 \in A$, which we know since $t^4 \in A$, we can take

$$t^{2k-2}t^3 = t^n \in A.$$

We do not, however, have $t \in A$ as

$$A = k \oplus kt^2 \oplus kt^3 \oplus \cdots$$

Proof of (b/c). The ideal (t^2, t^3) as an ideal of A is the set of

$$at^2 + bt^3,\ a, b \in A.$$

Furthermore,

$$k[t^2, t^3]/(t^2, t^3) \cong k$$

making it maximal.

As an ideal of B we get that

$$B/(t^2, t^3) \cong B/(t^2)$$

where $\bar{t}^2 = 0$ vanishing terms with t-degree greater than or equal to 2.

Explcitly

$$(t^2, t^3) = (t^2).$$

The inclusion $(t^2, t^3) \supseteq (t^2)$ is trivial.

For the other direction, since the ideal absorbs multiplication,

$$t \cdot t^2 = t^3 \in (t^2).$$

So the presence of t kills t^3. When quotienting we see that $\bar{t}^3 = 0$ is redundant since $\bar{t}^2 = 0$ anyways.

Proof of (d). We have t integral over A, and since

$$B = A[t]$$

we have that B is integral over A with generating set $\{1, t\}$.

The generating set is not a basis because it is not linearly independent.

We have

$$t \cdot t^2 - 1 \cdot t^3 = 0$$

so it is not linearly independent over A.

Problem (Product and subspace topologies)**.** Let $X = \mathbb{R} \times \mathbb{R}$ have the product topology, and let

$$P = \{(x, x^2) : x \in \mathbb{R}\}$$

have the subspace topology.

a) State the definition of the product topology and describe its basic open sets.

b) Let

$$U = \{(x, x^2) \in P : -1 < x < 1\}.$$

Prove that U is open in P. Is U open in X?

c) Prove that
$$f : \mathbb{R} \to P, \qquad f(t) = (t, t^2)$$
is continuous, using the universal property of the product topology and the subspace topology.

d) Construct the inverse of f and prove that f is a homeomorphism.

e) Determine whether P is closed in X, and justify your conclusion.

Proof of (a). The product topology, is the coarsest topology for which the universal property holds.

In particular, for continuous f on some space to a cartesian product, f is continuous if and only if
$$\pi_i \circ f$$
is continuous, where π_i denotes the i-th projection.

In the finite case, the basiselements are
$$\prod_{i=1}^{n} U_i,$$
where U_i is open in X_i.

In the infinite case the elements are
$$\prod_{\alpha} U_\alpha$$
where $U_\alpha = X_\alpha$ for all, but finitely many α. In the infinite case this differs from the box topology as the universal property can fail for the box topology. Take for instance $\mathbb{R}^\omega$ with the box topology. Define $f : \mathbb{R} \to \mathbb{R}^\omega$ by
$$t \mapsto (t, t, t, \ldots).$$
For each projection, this is simply the identity map which is continuous. However,
$$\prod_{n=1}^{\infty} (-1/n, 1/n)$$
is open in $\mathbb{R}^\omega$, but the inverse image is just $\{0\}$, which is not open.

■

Proof of (b). We have
$$U = P \cap (-1, 1) \times \mathbb{R}.$$

■

Proof of (c). I do not recall a universal property for the subspace topology.

Some open set in P is of the form

$$P \cap U$$

for open $U \subseteq X$.

Then

$$f^{-1}(P \cap U) = f^{-1}(P) \cap f^{-1}(U) = \mathbb{R} \cap f^{-1}(U) = f^{-1}(U).$$

We have

$$\begin{aligned} f^{-1}(U) &= f^{-1}\left(\bigcup_\alpha V_\alpha \times W\alpha\right) \\ &= \bigcup_\alpha f^{-1}(V_\alpha \times W_\alpha) \\ &= \bigcup_\alpha f_1^{-1}(V_\alpha) \cap f_2^{-1}(W_\alpha), \end{aligned}$$

where $f_i = \pi_i \circ f$.

We have $f_1(t) = t$ and $f_2(t) = t^2$. Thus these are continuous so the preimages are continuous. So we take the union of finite intersections of open sets which is open. This is slightly different from what is asked. We want to take basiselements $\bigcup_\alpha (-\varepsilon_\alpha, \varepsilon_\alpha) \times (-\delta_\alpha, \delta_\alpha)$ and from this clearly see that the inverse image is

$$\bigcup_\alpha (-\varepsilon_\alpha, \varepsilon_\alpha) \cap (-\sqrt{\delta_\alpha}, \sqrt{\delta_\alpha})$$

which clearly is open. ■

Proof of (d). All elements in P are of the form (t, t^2) so

$$f(t, t^2) = t$$

is clearly well-defined.

It is not hard to see that this is continuous either. ■

Proof of (e). The set is closed. Take the continuous $f(x, y) = y - x^2$. We have

$$f^{-1}(\{0\}) = P.$$

■